

Q15. DEMONSTRATE THE GIT COMMANDS TO CREATE A BRANCH, COMMIT CHANGES, AND MERGE IT TO MAIN

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ANSWER

Git is a powerful version control system that allows developers to manage and track changes in their codebase. The process of creating a branch, committing changes, and merging it into the main branch is a common workflow in Git. Here's how you can do it using Git commands:

1. Create a new branch

```
git checkout -b feature-branch
```

This command creates a new branch called feature-branch and switches to it. Branching allows you to work on a separate task without affecting the main code.

2. Make changes and stage them

After editing your files, add them to the staging area using:

```
git add .
```

This stages all the changed files to be committed.

3. Commit the changes

```
git commit -m "Added new feature in feature-branch"
```

This command saves the staged changes with a commit message.

4. Switch to the main branch

```
git checkout main
```

Before merging, go back to the main branch.

5. Merge the feature branch into main

```
git merge feature-branch
```

This command combines the changes from feature-branch into main.

6. (Optional) Delete the feature branch

```
git branch -d feature-branch
```

After merging, you can delete the branch to keep the repository clean.

This set of Git commands is essential in collaborative development to manage tasks, add new features, and keep the main code stable.